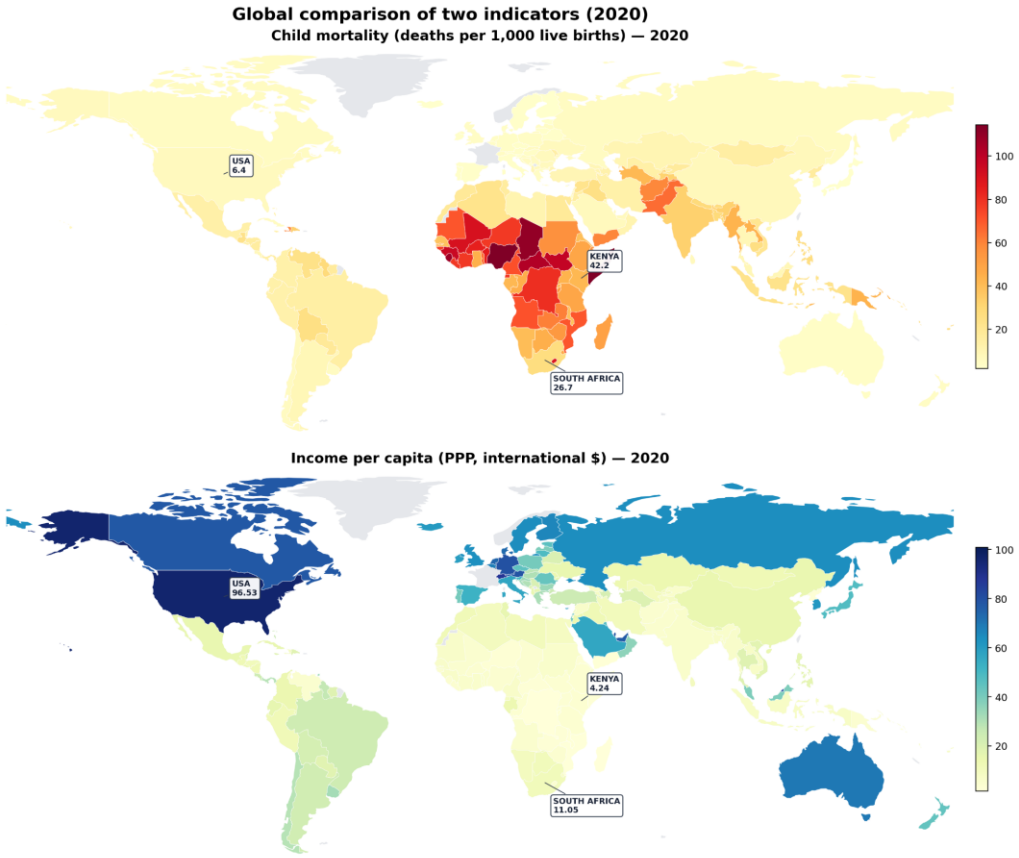
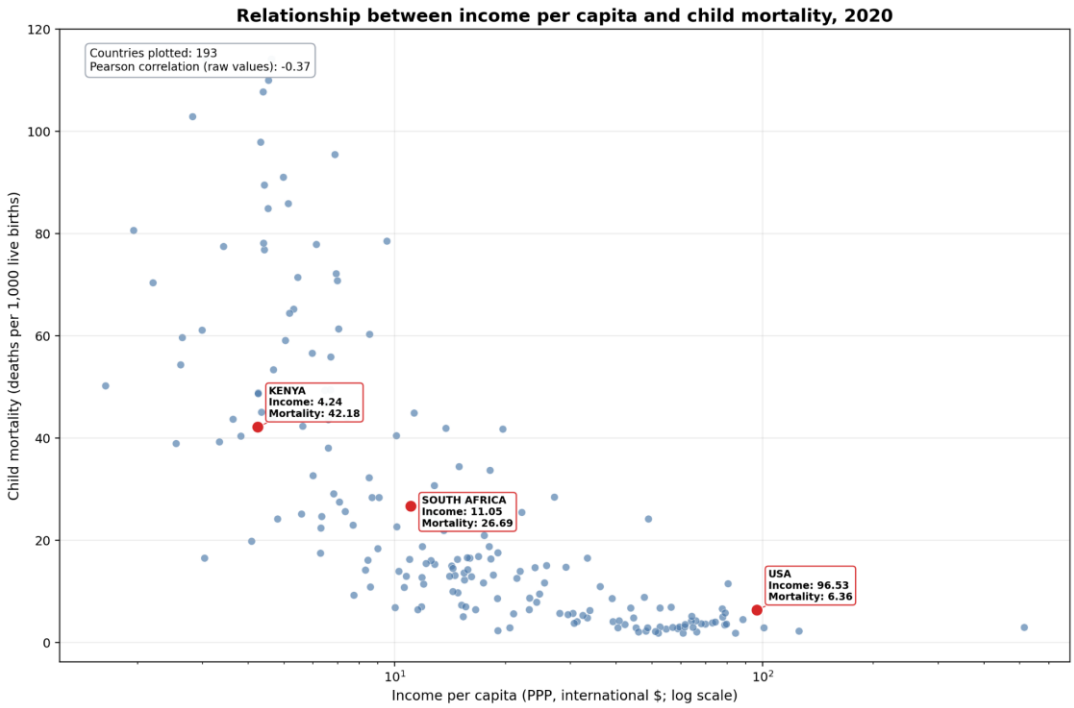
Exploring the Relationship Between Income per Capita and Under-Five Child Mortality Across Countries in 2020

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This world map illustrates the variation in under-five child mortality rates across countries in 2020, measured as deaths per 1,000 live births. The map highlights geographical differences in child survival, with higher mortality rates generally concentrated in lower-income regions, particularly in parts of sub-Saharan Africa. Kenya had a higher under-five mortality rate than South Africa and the United States, while the United States recorded a relatively low rate. These differences suggest disparities in child health outcomes and may reflect variations in healthcare access, socioeconomic conditions, nutrition, and maternal and child health services.

The second world map presents the geographical distribution of income per capita across countries in 2020, using the values reported in the dataset. The map demonstrates substantial differences in economic resources between countries. The United States had a considerably higher income per capita than Kenya and South Africa, while Kenya recorded the lowest value among the three countries. These economic differences may influence countries' capacity to finance healthcare services, improve living conditions, and support maternal and child health programs. However, national income per capita does not necessarily reflect how resources are distributed within a country.



The scatterplot examines the relationship between income per capita and under-five child mortality across countries in 2020. Each point represents a country, with income per capita on the horizontal axis and under-five mortality on the vertical axis. The overall pattern suggests a negative relationship: countries with higher income per capita generally tend to have lower under-five mortality rates. The United States appears among the countries with higher income and lower mortality, Kenya has lower income and higher mortality, and South Africa occupies an intermediate position. The estimated Pearson correlation of approximately −0.37 indicates a negative but relatively modest linear association in the matched data. This suggests that income is one factor associated with child survival, but other health system, social, and environmental factors also matter. Because this is an observational comparison, the results do not establish that higher income directly causes lower child mortality.

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3. Key findings

Country	Child mortality (per 1,000 live births)	Income per capita (PPP, dataset units)
USA	6.36	96.5305
Kenya	42.18	4.2380
South Africa	26.69	11.0524

The scatterplot shows a negative correlation of approximately −0.37 across the 193 countries with matched observations. This indicates that higher income per capita tends to be associated with lower child mortality, although the relationship is not especially strong. Correlation alone does not establish causation.

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1. Summary of Findings

The 2020 data suggest a negative relationship between income per capita and under-five child mortality across countries. Countries with higher income per capita generally tend to have lower child mortality rates, although there are exceptions. The United States had a relatively low child mortality rate and high income per capita, while Kenya had a higher child mortality rate and lower income per capita. South Africa fell between these two countries on both indicators. The scatterplot suggests that income per capita may be associated with child survival, but other factors, including healthcare access, maternal education, nutrition, and health system capacity, may also influence child mortality. These findings demonstrate an association rather than a causal relationship.

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2. Reflection on Working with AI

Working with AI made the data visualization process easier by helping organize the datasets, generate Python code, and create maps and a scatterplot. One of the most useful aspects was being able to visualize the relationship between the two indicators and highlight specific countries for comparison. It also reminded me of the importance of checking data quality, verifying results, and applying my own judgment when interpreting findings.